Chapter 1

Relativistic results

Chapter **??** ended with a number that undermines its own premise: a coherent speed of $0.47c$ to $0.87c$ at the ends of the certified windows. The next step was therefore to give the continuum solvers a relativistic electron and to run the HHS resonance again. The same step needed an independent relativistic particle code, because the question we want to answer, how large the electron heating of HHS is and what produces it, can only be asked of a code that has the noise HHS have. The chapter reports four calculations that share the set-up of Chapter **??**: an exact homogeneous relativistic driven oscillator; a relativistic grid Vlasov solver; a relativistic version of the Hermite solver; and a relativistic PIC. It then states what they agree on, which is the collapse of the mean-field response, and what they do not, which is the heating, by a factor that is at its smallest a factor of two and at its largest of order 10^2.

1.1 The exact relativistic driven oscillator

The role of relativity can be seen in a model small enough to be solved on the page. A spatially uniform plasma under a uniform drive responds only through its $k = 0$ moments, and for cold electrons with $q/m = -1$ in units $c = \omega_{pe} = 1$ the momentum per unit mass $u = \gamma v$ obeys $\dot{u} = -(\bar{E} + E_0 \cos \omega t)$ while the field obeys $\dot{\bar{E}} = -\bar{J} = nv$ up to the ion term. The exact relativistic kinetic version replaces v by the average of $u/\gamma(u)$ over the distribution, which depends on the whole distribution, not on a single mean; it is a closed set of ordinary differential equations for the distribution of u because a uniform field only translates it, and the exact solution is recorded to $\omega_{pe} t = 5000$ (`studies/relgrid/homog.json`).

The qualitative effect is easily derived. For a single oscillating electron without drive, $\mathrm{d}(\gamma \dot{x})/\mathrm{d}t = -x$, and to the first relativistic order $\gamma \simeq 1 + \dot{x}^2/2$ gives $\ddot{x}\,(1 + \frac{3}{2}\dot{x}^2) + x = 0$. With $x = A \cos \omega t$ the first harmonic of $\dot{x}^2 \ddot{x} = -A^3 \omega^4 \sin^2 \omega t \cos \omega t$ is $-\frac{1}{4} A^3 \omega^4 \cos \omega t$, so $\omega^2 + \frac{3}{8} A^2 \omega^4 = 1$ and

$$\omega \simeq \omega_{pe} \Big(1 - \tfrac{3}{16} \frac{v_m^2}{c^2} \Big), \qquad v_m = A\omega, \tag{1.1}$$

which a direct integration of the exact equation confirms: $(1-\omega)/v_m^2 = 0.1877$, 0.1884, 0.1911, 0.1956 for $v_m = 0.05$, 0.10, 0.20, 0.29 (script `note/check_relosc.py`, not a repository record). A heavier electron oscillates more slowly, and a drive held at the cold frequency ω_L walks out of phase with the response. Under resonant driving the amplitude is not constant but grows, $v_m \simeq v_q\, t/2$ with $v_q = v_q/v_{te}\, v_{te}$ the instantaneous drive in units of c; this is the HHS relation $v_q \simeq v_q^D \omega_{pe} t/2$ and the one that gave the nonrelativistic speeds of $0.47c$ at $\omega_{pe} t = 1000$ when $v_q = 9.5 \times 10^{-4}$. Substituting in Eq. (1.1) the accumulated phase mismatch is

$$\varphi(t) = \int_0^t \tfrac{3}{16} \Big(\frac{v_q t'}{2} \Big)^2 \mathrm{d}t' = \frac{v_q^2 t^3}{64}, \tag{1.2}$$

which reaches unity at $\omega_{pe}t = (64/v_q^2)^{1/3}$. For $v_q/v_{te} = 0.03$, $v_q^2 = 9.0\times10^{-7}$ and the estimate is $\omega_{pe}t \simeq 414$; the exact homogeneous kinetic ODE gives the time at which $W_{\rm ext}$ departs from the linear law by 10 % as $\omega_{pe}t = 392$, at which Eq. (1.2) gives 0.85. The estimate neglects thermal broadening, the ion mass and the change of the response when the electrons are no longer in step, and a 10 % departure is not exactly a phase of one radian; the agreement to 5 % should be read as a statement that the mechanism has the right order and the right dependence, $t_c \propto v_q^{-2/3}$, not as a derivation of the number.

The weak-drive case $v_q/v_{te} = 10^{-3}$ shows that the estimate is incomplete. The same formula gives $t_c \simeq 4000$ because $v_q^2 = 1.0\times10^{-9}$, whereas the exact homogeneous ODE departs at $\omega_{pe}t = 1429$ and the relativistic grid at 1309.5 (deterministic) or 1413.5 (noisy seeds). At that time the quiver speed is only $0.02c$ (about $0.65\,v_{te}$), too small for Eq. (1.1) to give a phase of order one. What takes over is the thermal shift: a Maxwellian of temperature $T = 10^{-3}m_ec^2$ has a relativistic mass correction of the order of T/m_ec^2, which shifts the plasma frequency by a relative amount of the same order, and a shift of 10^{-3} accumulates a radian of phase in 10^3 plasma periods. This estimate is of the right size and is consistent with the departure times of 1.3–1.4×10^3, but it was not derived with its coefficient and not tested by switching the thermal correction off; we regard it as a plausible reading, not a result. In the nonrelativistic grid with the same drive the work does not depart from the linear law at all up to $\omega_{pe}t = 3000$, while the mean-field energy is 0.71 of its linear value there (against 0.50 for the relativistic grid).

The relativistic law is, moreover, large. In the exact homogeneous problem the relativistic $W_{\rm ext}$ is 0.0056 of the nonrelativistic one at $\omega_{pe}t = 1000$ for $v_q/v_{te} = 0.03$, and a check in a relativistic code discriminates strongly: the nonrelativistic law misses $W_{\rm ext}$ by 38 % at $\omega_{pe}t = 500$ (record `studies/relhhs/gates_g1.json`). The relativistic mass increase of the coherently quivering electrons detunes the uniform resonance by itself, in a calculation with no fluctuations, no instability and no density modulation. That is a different mechanism from the one HHS describe, and the rest of the chapter asks what it leaves unexplained.

1.2 The relativistic grid

1.2.1 Construction and gates

The relativistic grid solver (`studies/relgrid.py`) integrates the one-dimensional, one-momentum-dimension Vlasov–Ampère system for electrons and ions on a fixed (x, u) grid, with $u = p/mc$ as the momentum variable, so that the force term is unchanged and only the streaming velocity u/γ, the current and the kinetic energy $(\gamma - 1)mc^2$ change. Space is spectral with N_x Fourier modes and the two-thirds mask, momentum uses 4096 nodes, and the step is $0.02\,\omega_{pe}^{-1}$ with an iterative or Taylor midpoint. It has no Hermite truncation and no closure, a property that matters: it can lose positivity, but it cannot develop the truncation instability of Chapter **??**. Seeds are either deterministic, like those of the Hermite code, or built from shot noise of a stated particle number per cell.

The gates were declared in `studies/relgrid_gates.py` before any run. The nonrelativistic limit reproduces the earlier nonrelativistic record to 1.5×10^{-10} (gate $\le 10^{-9}$) and the laboratory and oscillation frames agree to 1.5×10^{-10}. The homogeneous problem agrees with the exact kinetic ODE with a second-order error (the Δt^2 ratio is 3.99) and Richardson-extrapolated errors of 5.6×10^{-7} in $\bar{E}$, 5.2×10^{-7} in U_e, 3.4×10^{-7} in W and 7×10^{-8} rad in phase. Three declared checks failed. The Landau-root check C fitted the real part of E_1, which for a real density seed is noise, and failed; the replacement C2, declared before it was run and using the imaginary part, passes with errors of 8.9×10^{-5} in frequency and 4.5×10^{-4} in damping at $k\lambda_{De} = 0.3, 0.4$, closer to the relativistic roots ($1.15849 - 0.01203i$ at $k\lambda_{De} = 0.3$) than to the nonrelativistic ones ($1.16001 - 0.01260i$). The ledger gate D, a defect of at most 10^{-4} of the work, failed: 2.3×10^{-4} for the deterministic run at $\Delta t = 0.02$, with 15 of 16 runs

above 10^{-4}. Its companion D2, that the defect should fall by a factor of 3–5 when Δt is halved, also failed with a ratio of 8.0 (the value at $\Delta t = 0.01$, 2.9×10^{-5}, does pass). The grid ledger is therefore weaker than that of the Hermite code by about four orders of magnitude, and this is carried into every grid result below. Refinement pairs in N_x, momentum nodes and time step agree to $\omega_{pe}t = 1000$ under the 10 % rule.

1.2.2 Collapse times

For the deterministic seed at $v_q/v_{te} = 0.03$, $N_x = 32$, the work leaves the linear law by 10 % at $\omega_{pe}t = 392$. The mean-field energy relative to its linear value is 0.854 at $\omega_{pe}t = 500$ and 0.041 at 1000, which corresponds to $3.7\,E_{\rm th}$ (that is $1.9 \times 10^{-3} n_e m_e c^2$) of absolute mean-field energy at 1000. HHS show, by our reading of the logarithmic plot, a mean-field energy of about 10^{-3} near $\omega_{pe}t \simeq 800$ and below 10^{-4} at 1000 in the same units; the timing of the relativistic collapse is consistent with theirs, the late level of the deterministic grid is about a factor 20 above theirs. With shot-noise-like seeds the collapse time is 367–392.5, depending on the particle number of the seed and on N_x, among the seeds that stay valid long enough to define it; the nonrelativistic grids with the same noise collapse at 769.5–866.5. The deterministic nonrelativistic Hermite code shows no collapse in its windows. At $v_q/v_{te} = 10^{-3}$ the relativistic grid departs at 1309.5 (deterministic) and 1413.5 (shot noise), and its mean-field energy is 0.50 of the linear value at $\omega_{pe}t = 3000$, with a heating of $0.09\,E_{\rm th}$ (nonrelativistic: 0.71 and below $0.3\,E_{\rm th}$). Relativity is thus both necessary and sufficient for a collapse before $\omega_{pe}t \simeq 800$ in the continuum models: necessary because the nonrelativistic ones do not collapse before 770 even with noise, and sufficient because the deterministic relativistic grid collapses at 392 with no instability at all. This is the first of the two results to which the rest of the note returns.

1.2.3 Shot-noise runs and the loss of validity

The HHS heating of $50\,E_{\rm th}$ is the second result to be reproduced, and the deterministic grid does not do it: the electron energy gain in the HHS measure is $0.018\,E_{\rm th}$ at $\omega_{pe}t = 500$ and $0.056\,E_{\rm th}$ at 1000, with the minimum of f equal to -3×10^{-12}, so that nothing is wrong numerically. Detuning switches off the mean field, and it does not heat. The remaining candidate is a fluctuation level. The grid was therefore run with density seeds drawn from the shot noise of $N_{\rm pc}$ particles per cell. For $N_{\rm pc} = 205$, the PIC particle number, the run is invalid: $f < -10^{-3}$ already within ω_{pe}^{-1} of the start, the noise amplitude of the lowest modes being comparable to the thermal energy; it is kept and labelled so. For $N_{\rm pc} = 2.05 \times 10^5$ (a seed of $U_{k0} = 7 \times 10^{-7}$) the heating begins at about $\omega_{pe}t = 600$ and the values at 1000 are $15\,E_{\rm th}$ ($N_x = 32$), 51 ($N_x = 64$) and 30 ($N_x = 128$), with 50 for $N_{\rm pc} = 2.05 \times 10^4$ at $N_x = 32$ and 209 for the nonrelativistic control. These numbers are of the HHS size and suggestively so, and they are not results. In all of these runs $f < -10^{-3}$ appears between $\omega_{pe}t = 240.5$ and 407.5, the box edge fraction exceeds 10^{-6} between 350 and 500, and the ledger exceeds 10^{-4} between 350 and 510. The heating at $\omega_{pe}t = 500$ is 5–10 $E_{\rm th}$ for $N_x = 64, 128$ and the valid window ($t < 350$) has less than $0.01\,E_{\rm th}$. The question whether the cascade at the high k resolved by $N_x = 64$ and 128 would heat if the code could survive remained open. The answer from the grid is therefore: the heating of HHS is not reproduced within validity, and cannot be assessed beyond it.

A finer Fourier range was tried in the nonrelativistic grid (the `hkr` study), reaching $N_x = 1024$ and $k\lambda_{De} = 1.69$, since HHS' modulational and decay instabilities populate modes up to the inverse Debye length. The deterministic gain at $\omega_{pe}t = 1000$ is $2.3\,E_{\rm th}$ for $N_x \geq 512$ (the chains at $N_x = 256, 640, 1024$ agree over their certified windows), about 20 times smaller than the HHS value. With shot noise the grid loses positivity at $\omega_{pe}t = 69$–163.5 and has gained $0.013\,E_{\rm th}$ by then, whatever N_x, so the Fourier range alone does not explain the heating (declared question Q1 of the study). Five of the validation checks of the study at $k\lambda_{De} = 1$ failed

as specified and are recorded in `studies/hkr/validation*.json` together with the follow-ups that were declared after each.

1.3 The relativistic Hermite model

1.3.1 The model and its gates

The Hermite model was extended in two steps. In the variable $u = p/mc$ the force term is unchanged and only the streaming, the current and the kinetic energy involve $v(u) = u/\gamma$. The velocity operator $V = P\,\mathrm{diag}(v(u_0 + a\lambda_j))P^T$ is diagonalized once in the Hermite basis, with the same eigendecomposition as the exact Lawson streaming factor of the nonrelativistic code, so that streaming and the current remain matrix functions of one decomposition. A relativistic option was then added to the parent solver (SPECTRAX PR 88) and the pump frame was extended to it, at the pin `6a6606b` of `integration/dark-baseline-2`. The field-scaled closure is a diagonal damping outside the kernel and is unchanged, and it remains exact in the Lawson factor. Three things are not supported and are refused by the code: the exact streaming factor in the relativistic case (streaming is explicit), Dougherty collisions, and the magnetic term in more than one transverse axis (the three-velocity case couples axes through γ and would cost $O(N^4)$). Nonrelativistic runs are bitwise unchanged.

The gates were again declared first. In a limit with $c \to \infty$ the right-hand side agrees with the nonrelativistic solver to 10^{-15}; a single gyrating electron matches the exact relativistic frequency to 2×10^{-10}; the magnetic force does no work to 3×10^{-15}; and the ledger including $(\gamma - 1)mc^2$ closes to 6.6×10^{-10} for an electron driven to $0.13c$ and to 8×10^{-9} at $0.83c$. The spectrum check c failed as declared, because the discrete spectrum of a truncated Hermite matrix is undamped and the control failed in the same way (the declaration was wrong); the replacement c2 on the response function passes at 3×10^{-5}. The nonlinear 1D2V run with a transverse field failed at $\omega_{pe}t \simeq 8$ with a NaN from the truncation instability, which is not a relativistic effect (the frozen-field growth is 4.7 relativistic against 4.9 nonrelativistic); the variant with a hypercollision of strength 10, declared afterwards, closes at 6.8×10^{-6} and removes 0.7 % of the energy change at a mean speed of $0.59c$.

The production study then ran the HHS resonance in eight configurations, $N_x \in \{32, 64\} \times N_n \in \{64, 128\} \times c \in \{16, 64\}$, to $\omega_{pe}t = 1000$. The declared gates all pass. G1, the homogeneous limit against the exact kinetic ODE for $\omega_{pe}t \le 500$ at $N_n = 64, 128$: 1.7×10^{-9} and 2.5×10^{-9} in $\bar{E}$, 5.5×10^{-10} and 7.8×10^{-10} in W_{ext}, 1.6×10^{-9} and 2.3×10^{-9} rad in phase, against a threshold of 10^{-6}; the nonrelativistic ODE misses W_{ext} by 38 % as noted. G2, that 12 refinement pairs agree to 10 % in ΔK_e and W_{ext} to a declared 400, passes to 1000. G3, positivity, and G4, the ledger including the rest-mass term to 10^{-4}, pass with a maximum of 1.8×10^{-8} for $t \le 400$. The fraction of the energy in the top third of the Hermite ladder is below 3×10^{-8}.

1.3.2 Collapse and heating

The work leaves the linear law by 10 % at $\omega_{pe}t = 392.5$ in all eight runs, to the resolution of the sampling, against 392 for the grid. The mean-field energy relative to the linear law is 0.855 at 500 (grid 0.854) and 0.0409 at 1000 (grid 0.0407). The HHS measure of the thermal gain is $0.0177\,E_{\mathrm{th}}$ at 500 and $0.056\,E_{\mathrm{th}}$ at 1000, identical in all eight runs to three digits and identical to the deterministic grid (0.0559). Two independent discretizations, one with a closure and one without, agree on the mean field and on the heating to the digits shown. This is the strongest evidence in the note that the continuum answer to the deterministic problem is right, and it is also the evidence that the deterministic problem does not heat.

Two follow-ups were declared after the production gates and both failed. F1, certification to $\omega_{pe}t = 2000$, fails: $f_e < -10^{-3}$ appears at 1528 and 1584.5 for the two base runs, the

ledger exceeds 10^{-4} at 1855 for $N_x = 32$, and the $N_x = 64$ run was stopped at 1800 once the horizon had been fixed; the pairs do not disagree. The heating stays below $0.7\,E_{\rm th}$ throughout. The model is therefore certified to about $\omega_{pe}t = 1530$–1585, past the validity of the shot grid ($t < 350$) and short of the declared 2000. F2, shot-noise seeds with the recipe of the grid, fails: with $N_{\rm pc} = 103$, the PIC value, $N_x = 32$, $N_n = 64$ has $f_e < -10^{-3}$ at $\omega_{pe}t = 0.5$, the ledger above 10^{-4} at 10.5 and a positivity stop at 143, and the $N_x = 64$, $N_n = 128$ run never finished its first 100-unit chunk and was killed after 70 minutes; with $N_{\rm pc} = 2.05 \times 10^5$ positivity fails at 367 ($N_x = 32$) and 265 ($N_x = 64$), with stops at 660 and 380, against 246–278 for the grid. The Hermite model reproduces the deterministic relativistic collapse to three or four digits and a ledger of 10^{-8}, and cannot reach the PIC regime before losing positivity. The cost of a $T = 1000$ run is 2–19 minutes on one thread under load.

1.4 The relativistic PIC

1.4.1 Set-up

The independent test is a particle code. Dark-JAX-in-Cell (commit `d547579`) on JAX-in-Cell (`2d693cb`) has a relativistic Boris pusher of the Vay family. The configuration reproduces what HHS state about their own: 1000 cells of $0.04\,c/\omega_{pe}$ in a box $L = 40\,c/\omega_{pe}$, $m_i/m_e = 1836$, $T_e = T_i = 10^{-3} m_e c^2$, a uniform prescribed drive with $v_q/v_{te} = 0.03$ at the cold frequency, quadratic shape functions and $\Delta t = 0.02\,\omega_{pe}^{-1}$; the same code with the Newtonian momentum is the control. Before any production run the pusher was tested on single particles (S1–S3: pure electric and pure magnetic motion exact, the $\mathbf{E} \times \mathbf{B}$ drift error 1.7×10^{-4} and second order), and in a pilot to $\omega_{pe}t = 100$ the Gauss-law and continuity residuals were at 10^{-14}. The first twelve production runs (three seeds each, to $\omega_{pe}t = 2000$) passed G1, G2, G3, G6, G7 (quintic shape) and G9 (no-drive control), and failed G4 (the heating at 412 particles per cell was 19 % above that at 103), G5 (a cell of 0.02 raised it by 19 % and 18 % at 400 and 600) and G10 (the $N_x = 32$ Hermite record lies one to two orders of magnitude below the PIC). These failures prompted the convergence ladder rather than a conclusion.

1.4.2 The convergence ladder

The ladder was declared in `studies/relpic/conv_gates.py` (follow-up in `conv_followup_gates.py`) and run on CPU with 15 new runs. Its results for the HHS measure at $\omega_{pe}t = 1000$ in units of $E_{\rm th}$ are given in Table 1.1. In particles per cell at $\Delta x = 0.04$ the heating is 21.5 ± 0.6 (103), 23.7 (206), 24.6 ± 2.0 (412, five seeds), 25.0 (824) and 24.9 (1648): a rise of about 15 % between 103 and 412 and then a plateau with successive changes of 1.4 % and 0.1 %. Halving the cell at 103 particles per cell gives 24.1 (0.02) and 24.3 (0.01), a change of 0.8 % between the two. Quintic shape functions at 412 give 25.7, and halving the time step at 103 gives 21.1 against 21.5. The gates C1 to C4 (successive differences within 5 % or twice the seed spread) pass; their tolerances contain an allowance of $2\sqrt{2}$ times the seed standard deviation, which at 412 particles per cell is $2.0\,E_{\rm th}$. At $\omega_{pe}t = 2000$ and 412 particles per cell the heating is 28.0 ± 3.7. The plateau is $\simeq 25\,E_{\rm th}$, half of the re-digitized HHS value of 50.1 (52.7 at 2000).

1.4.3 Newtonian control, mirrored loading, extrapolation

The Newtonian control gives 55.9 at 103 particles per cell (three seeds) and 62.7 ± 9.4 at 412 (two seeds), 68.9 at $\omega_{pe}t = 2000$. It is 1.25 times the HHS value at 1000 and 2.5 times the relativistic one; the relativistic-to-Newtonian ratio is 0.39 at 1000 and 0.41 at 2000 (9.5 standard errors from unity at the latter). Relativity therefore reduces the heating by a factor of about 2.5 in the same code, a statement that does not depend on any comparison with HHS. A dependence on noise was tested next. Velocities were loaded in exactly cancelling $(v, -v)$ pairs

Table 1.1: PIC electron heating in the HHS measure, $\Delta E_e/E_{\rm th}$ at $\omega_{pe}t = 1000$ (trailing $200\,\omega_{pe}^{-1}$ mean), $v_q/v_{te} = 0.03$, relativistic unless stated, quadratic shape, $\Delta x = 0.04$, $\Delta t = 0.02$ except where varied (record `studies/relpic/conv_summary.json`). The $\pm$ is the standard deviation over seeds.

variation	$\omega_{pe}t = 1000$
103 ppc (3 seeds)	21.5 ± 0.6
206 ppc	23.7
412 ppc (5 seeds)	24.6 ± 2.0
824 ppc	25.0
1648 ppc	24.9
$\Delta x = 0.02$, 0.01 at 103 ppc	24.1, 24.3
quintic shape at 412 ppc	25.7
$\Delta t = 0.01$ at 103 ppc	21.1
mirrored $(v, -v)$ loading, 103 ppc (2 seeds)	21.9, 22.6
mirrored loading, 206 ppc	23.4
Newtonian, 103 ppc (3 seeds)	55.9
Newtonian, 412 ppc (2 seeds)	62.7 ± 9.4
HHS, Fig. 2 re-digitized	50.1

on adjacent particles, which removes the sampled long-wavelength current noise; the heating is unchanged (21.9 and 22.6 at 103 particles per cell against 21.5 ± 0.6, and 23.4 at 206 against 23.7) and does not decrease with particle number. The plateau is therefore not a noise level falling as $1/N_{\rm ppc}$. The test cancels only the long-wavelength current noise, and says nothing about the density noise at the scale of a cell.

The declared extrapolation to infinite particle number and zero cell size (C5), multiplying the 412 value by fitted ratios in $N_{\rm ppc}^{-1}$, $N_{\rm ppc}^{-1/2}$ and Δx^2, gives $30.1 \pm 3.8\,E_{\rm th}$. We regard it as an overcount, because the particle-number and cell-size corrections are the same 15 % rise between 103 and 412 particles per cell seen through two ladders and the combined point (412 particles per cell with $\Delta x \leq 0.02$) was not run, so the product counts one effect twice. The plateau reading, $\simeq 25$, is the one used. The declared comparison with HHS (C6: agreement within 10 % including the digitization uncertainty) fails, with the relativistic plateau at 0.50 of HHS (0.60 for the extrapolated value) and the Newtonian result at 1.25. Both failures are recorded as declared.

1.4.4 Mean field and collapse in the PIC

The PIC collapses earlier than the deterministic continuum models: the work departs from the linear law by 10 % at $\omega_{pe}t = 273$–286 relativistic and 292–311.5 Newtonian (412 particles per cell), against 392 for the deterministic relativistic continuum. The noise already suffices to depart from the linear law by 10 % with the Newtonian pusher, so in the PIC the timing mixes relativity with noise, and the statement that relativity sets the timing is made in the deterministic models. At later times relativity matters: the mean-field energy relative to the linear law at $\omega_{pe}t = 800$ is 0.008 relativistic against 0.076 Newtonian, and at 1000 it is 0.003 in both. The absolute mean-field energy retained at 1000 is $0.25\,E_{\rm th}$ in the relativistic PIC (against $3.7\,E_{\rm th}$ in the deterministic grid), peaks at $11\,E_{\rm th}$, and the $k \neq 0$ field energy peaks at 2–$6\,E_{\rm th}$. The PIC mean field therefore falls much further than the continuum one, and in this respect the PIC is closer to HHS, though both still depend on the noise.

1.5 The factor of two with HHS

The relativistic PIC plateau of $25\,E_{\mathrm{th}}$ is half of the HHS value, after a correction of our own comparison number that is described in Chapter **??**. We do not know why, and we have not tested most of the candidate causes. The candidates, in the order in which we consider them worth testing, are these. The first is the shape function and the scheme: SHARP uses fifth-order splines in deposition and in the interpolation of the force, our baseline uses quadratic functions, and the quintic test in our code changed the plateau by only 4 % (25.7 against 24.6), which argues against a large effect but does not replace running the actual scheme. The second is the particle number and loading: HHS state a range of 2×10^{2} to 2×10^{5} particles per cell and show, to our knowledge, one curve for each case, so it is not known which number produced the plotted curve; we did not run the combined point of high particle number and small cell, and the heating rises with particle number, with no fall as $1/N_{\mathrm{ppc}}$, in a way that the mirrored-loading test does not explain. The third is the pusher: ours is a Boris/Vay-family implementation and theirs is the Vay algorithm, which differ in the treatment of $\mathbf{E}\times\mathbf{B}$, a distinction that should not matter in one dimension without a magnetic field but is not excluded for a resonant relativistic case. The fourth is the plotted energy: our $k\neq0$ one-dimensional electron variance with the box-mean flow removed is our reading of their definition, and it is conceivable that the HHS curve includes something we exclude, for example a contribution from the coherent $k=0$ motion at finite γ or energy already in $k\neq0$ waves that is counted differently. The last candidate is the digitization itself, about 3 %, which cannot account for the factor.

None of these has the form of a clean test, so none is a result. We emphasize that the Newtonian code gives a value above HHS, so the relativistic result alone does not suggest that something is missing from the physics of the PIC; the discrepancy is a factor within the range spanned by the physical and numerical variations, a factor that a fifth-order shape, a larger particle number or a different treatment of the high-k cutoff could, in principle, supply. We make no claim that HHS or our PIC is wrong.

1.6 The continuum–PIC gap

The second discrepancy is much larger. The relativistic PIC plateau is 445 times the continuum heating at $\omega_{pe}t=1000$ (24.9 at 1648 particles per cell against 0.056), and at 2000 the continuum remains below $0.7\,E_{\mathrm{th}}$ while the PIC is 28.0 ± 3.7. The deterministic grid and Hermite codes agree to three digits and the Hermite one satisfies a ledger of 10^{-8} while remaining positive to 1530–1585; there is no indication that the continuum value is numerically wrong. The PIC, for its part, passes its own checks over a ladder that spans a factor of 16 in particle number and of 4 in cell size. Both statements can be true because the two models solve the same equations with different initial conditions: the deterministic seed has a smooth density modulation on the lowest few modes of relative amplitude 5×10^{-4}, while the PIC has the broadband shot noise of a finite number of particles, with energy at all k up to the cell scale. The continuum codes with noise lose positivity (Sections 1.2 and 1.3) before the heating develops. The mirrored loading test and the plateau argue that the seed of the PIC heating is not the long-wavelength current noise and does not decrease as noise is reduced by sampling. What is left, as a hypothesis and not a finding, is that the heating is seeded by fluctuations at or near the cell scale, which the continuum codes cannot represent at their Fourier cut-off and which the mirrored loading does not remove. The consequences of the hypothesis are developed in Chapter **??**. Here we record only its precise form: the two descriptions agree on the detuning and disagree on the heating by a factor 445, and the disagreement is not a numerical convergence failure of either side as far as the checks we ran can tell.

Table 1.2 collects the comparison, and Fig. 1.1 reproduces the paper figure of the mean field and the heating.

Table 1.2: Resonant drive $v_q/v_{te} = 0.03$ across models: collapse time of W_{ext} (10 % departure from the nonrelativistic linear law), mean-field energy over its linear value at $\omega_{pe}t = 1000$, HHS-measure heating at 1000, and the time to which the model is valid. †Shot-noise seeds: the values are at 1000 but the runs lose positivity earlier, so they are not results; ranges are over particle numbers 2×10^4–2×10^5 per cell and $N_x = 32$–128. PIC: seed mean at 412 particles per cell (five seeds relativistic, two Newtonian). The Hermite row is $N_x = 32$, $N_n = 64$. HHS: read from their Fig. 2 and SM.

model	collapse	$\bar{E}^2/2$ over lin.	$\Delta E_e/E_{\text{th}}$	valid to
Hermite, nonrel.	none (≤ 1000)	0.96	2.5	1000
grid, nonrel., shot†	769.5–866.5	0.12–0.26	99–209	240–394
grid, rel., det.	392	0.041	0.056	1000
Hermite, rel., det.	392.5	0.041	0.056	1528–1585
grid, rel., shot†	367–392.5	0.008–0.30	15–51	240.5–407.5
PIC, rel.	273–286	0.003	24.6	—
PIC, Newtonian	292–311.5	0.003	62.7	—
HHS (digitized)	~ 800	—	50.1	—

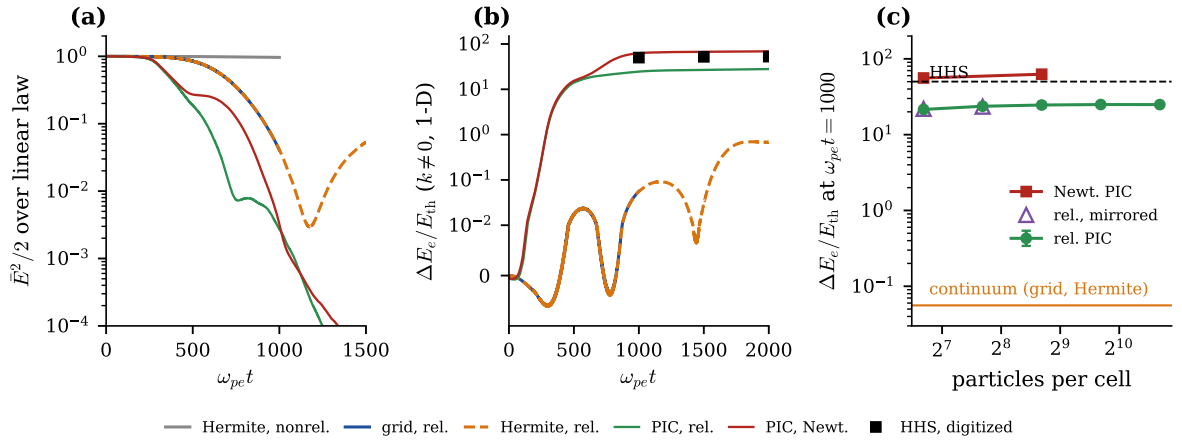


Figure 1.1: Resonant drive $v_q/v_{te} = 0.03$ (figure of the paper). (a) Mean-field energy over the exact linear law (trailing $200\,\omega_{pe}^{-1}$ mean) for the nonrelativistic Hermite model, the relativistic grid, the relativistic Hermite model and the relativistic and Newtonian PIC. (b) HHS measure of the $k \neq 0$ electron energy change over E_{th} (symmetric logarithmic scale); squares: HHS Fig. 2 re-digitized. The continuum curves coincide to $\omega_{pe}t = 1000$ and remain below $\sim 1\,E_{\text{th}}$ to 2000. (c) Heating at $\omega_{pe}t = 1000$ against particles per cell, with the relativistic plateau, the Newtonian PIC, mirrored loading, the HHS value and the continuum value.

Bibliography